

# SEMIGROUP NOTES

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## 1. PRELIMINARIES

Let's begin by recalling some basic definitions and fixing terminology.

In all of what follows,

- $E$  denotes a real Banach space,
- $\mathcal{L}(E)$  is the set of bounded linear operators on  $E$ ,
- $E'$  represents the dual space (i.e., the set of bounded linear functionals on  $E$ ),
- $\|\cdot\|$  denotes either the norm on  $E$  or the operator norm on  $\mathcal{L}(E)$ , depending on context.

Given  $R \subset \mathbb{R}$ ,

- $C_0(R, E)$  is the set of continuous functions from  $R$  to  $E$  and
- $C_1(R, E)$  is the set of continuously differentiable functions from  $R$  to  $E$ .

The notion of derivative used in the last definition is discussed below.

A (not necessarily bounded) linear operator  $A$  from  $\mathcal{D}(A) \subset E$  into  $E$  is called **closed** if, for each sequence  $(u_n) \subset \mathcal{D}(A)$  such that  $u_n \rightarrow u$  for some  $u \in E$  and  $Au_n \rightarrow y$  for some  $y \in E$ , we have  $u \in \mathcal{D}(A)$  and, in addition,  $y = Au$ .

Let  $A: \mathcal{D}(A) \rightarrow E$  be a closed linear operator and let  $I$  be the identity map on  $E$ . The set  $\rho(A)$  of all  $\lambda \in \mathbb{C}$  such that  $\lambda - A := \lambda I - A$  is a bijection from  $\mathcal{D}(A)$  to  $E$  is called the **resolvent set** of  $A$ . Its complement  $\mathbb{C} \setminus \rho(A)$  is called the **spectrum**. Given  $\lambda \in \rho(A)$ , the **resolvent operator** of  $A$  is defined by

$$R_\lambda(A)u = (\lambda - A)^{-1}u \quad (u \in E).$$

## 2. LINEAR DIFFERENTIAL EQUATIONS AND EXPONENTIAL PATHS

This section contains a brief review of linear ordinary differential equations (ODEs) in Banach space when the linear operator that defines the ODE is a bounded linear operator.

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To set the stage, recall that an (autonomous) linear ODE in  $\mathbb{R}^n$  has the form

$$\dot{x}(t) = Ax(t)$$

where  $x(t)$  is an  $n$ -vector for each  $t$  and  $A$  is  $n \times n$ . The matrix  $A$  is sometimes called the “vector field.” A first step to extend these concepts to an abstract Banach space  $E$  is to view  $t \mapsto x(t)$  as a differentiable function from  $\mathbb{R}_+$  to  $E$  and  $A$  is a *bounded* linear operator from the space to itself. (This includes the finite-dimensional case, since all linear operators on finite-dimensional spaces are bounded. The derivative  $\dot{x}$  is defined below.)

It turns out that the boundedness restriction leaves out many interesting and important systems. On the other, it illustrates the ideal case, and highlights some of the properties we hope to discover for the solutions of general ODEs (driven by possibly unbounded linear operators) in Banach space.

**2.1. Calculus.** We recall some facts concerning calculus for functions mapping a subset of the real line into a Banach space. To this end, let  $u$  be an element of  $C_0(\mathbb{R}_+, E)$ . If, for some  $t > 0$ , the limit

$$\dot{u}(t) := \frac{d}{dt}u(t) := \lim_{h \rightarrow 0} \frac{\|u(t+h) - u(t)\|}{h}$$

exists, then we say that  $u$  is *differentiable* at  $t$  and call  $\dot{u}(t)$  the *derivative* of  $u$  at  $t$ . In the case where  $t = 0$  we replace  $h \rightarrow 0$  with  $h \downarrow 0$ . We say that  $u$  is *differentiable* on  $\mathbb{R}_+$  if  $u$  is differentiable at  $t$  for all  $t \in \mathbb{R}_+$ . We say that  $u$  is *continuously differentiable* on  $\mathbb{R}_+$  and write  $u \in C_1(\mathbb{R}_+, E)$  if, in addition,  $\dot{u} \in C_0(\mathbb{R}_+, E)$ .

Given  $u \in C_0(\mathbb{R}_+, E)$ , the (Riemann) *integral* of  $u$  over  $[a, b]$  is the unique point  $I \in E$  such that

$$\int_a^b \langle u(s), u' \rangle ds = \langle I, u' \rangle \quad \text{for all } u' \in E', \quad (1)$$

where the left hand side is an ordinary Riemann integral for real-valued functions. We write  $I = \int_a^b u(s) ds$ , so that  $\int_a^b \langle u(s), u' \rangle ds = \left\langle \int_a^b u(s) ds, u' \right\rangle$  for all  $u' \in E$ . A proof of existence and uniqueness, as well as the properties listed below, can be found in Section 5.1.2 of [Bühler and Salamon \(2018\)](#).

**Lemma 2.1.** *If  $u \in C_0(\mathbb{R}_+, E)$  and  $a, b, c \in \mathbb{R}_+$  with  $a \leq b \leq c$ , then*

- (i)  $\int_a^c u(s) ds = \int_a^b u(s) ds + \int_b^c u(s) ds$ ,
- (ii)  $\int_a^b u(s+r) ds = \int_{a+r}^{b+r} u(s) ds$  whenever  $r \geq 0$ ,
- (iii)  $\|\int_a^b u(s) ds\| \leq \int_a^b \|u(s)\| ds$ ,

(iv)  $u(t) = \lim_{h \rightarrow 0} (1/h) \int_t^{t+h} u(s) \, ds$  for all  $t \in \mathbb{R}_+$ , and

(v)  $L \in \mathcal{L}(E)$  implies  $L \int_a^b u(s) \, ds = \int_a^b Lu(s) \, ds$ .

If, in addition,  $u$  is continuously differentiable on  $\mathbb{R}_+$  with derivative  $\dot{u}$ , then

$$u(b) = u(a) + \int_a^b \dot{u}(s) \, ds. \quad (2)$$

**2.2. Exponentials of Linear Operators.** The scalar exponential of  $a \in \mathbb{R}$  can be defined by  $\exp(a) := e^a := \sum_{n=0}^{\infty} a^n / (n!)$ . Analogously, the **exponential** of  $A \in \mathcal{L}(E)$  is given by

$$\exp(A) := e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}. \quad (3)$$

**Lemma 2.2.** *If  $A, B \in \mathcal{L}(E)$ , then*

- (i)  $\exp(A)$  is a well-defined element of  $\mathcal{L}(E)$  and  $\|\exp(A)\| \leq \exp(\|A\|)$ .
- (ii) If  $AB = BA$ , then  $\exp(A + B) = \exp(A) \exp(B)$ .
- (iii)  $\exp(0) = I$ .

It is immediate from (ii) that if  $m$  is any positive integer, then  $\exp(mA) = (\exp(A))^m$ .

The next lemma considers exponential flows in  $\mathcal{L}(E)$ .

**Lemma 2.3.** *Given  $A \in \mathcal{L}(E)$ , the function  $\varphi(t) := \exp(tA)$  is continuously differentiable in  $t$ , with*

$$\dot{\varphi}(t) = A\varphi(t) = \varphi(t)A \quad (t \in \mathbb{R}). \quad (4)$$

*Proof.* Fix  $A \in \mathcal{L}(E)$ . Applying (ii) of Lemma 2.2 yields

$$\frac{\varphi(t+h) - \varphi(t)}{h} = \frac{\exp(tA + hA) - \exp(tA)}{h} = \exp(tA) \frac{\exp(hA) - I}{h}$$

Using the definition (3) and taking the limit as  $h \downarrow 0$  gives  $\dot{\varphi}(t) = \varphi(t)A$ . A small variation on the argument shows that  $\dot{\varphi}(t) = A\varphi(t)$  also holds.  $\square$

Lemma 2.1 and Lemma 2.3 together yield

$$\exp(tA) = \exp(sA) + A \int_s^t \exp(rA) \, dr \quad \text{for all } s \leq t. \quad (5)$$

**2.3. Initial Value Problems.** Fix  $A \in \mathcal{L}(E)$  and consider the **initial value problem** (IVP)

$$\dot{u}(t) = Au(t) \quad \text{for all } t \geq 0 \text{ with } u(0) = u_0 \in E \text{ given.} \quad (6)$$

A solution to this problem is a differentiable  $u \in C_0(\mathbb{R}_+, E)$  that obeys (6).

**Proposition 2.4.** *If  $A \in \mathcal{L}(E)$ , then  $u$  defined by  $u(t) = \exp(tA)u_0$  is the unique solution to the IVP (6).*

That  $u$  solves the IVP is immediate from the properties in Lemmas 2.2–2.3. We prove uniqueness in a more general setting below.

Proposition 2.4 gives a complete description of the solution of ODEs in Banach space when the vector field  $A$  is a bounded linear operator. As well as studying flows of the form  $t \mapsto \exp(tA)u_0$  in  $E$  we can also directly analyze the dynamics of flows  $t \mapsto \exp(tA)$  in  $\mathcal{L}(E)$ . We notice that these flows are continuous (in fact differentiable, by Lemma 2.3) and obey  $\exp((s+t)A) = \exp(tA)\exp(sA)$  for all  $s, t \geq 0$ . The latter property is called the *semigroup* property of the flow.

In order to generalize these ideas and handle ODEs with unbounded vector fields, we now study a more general notion of continuous semigroups in  $\mathcal{L}(E)$ .

### 3. INTRODUCTION TO SEMIGROUPS

Add roadmap.

**3.1. Definitions.** Let  $(S_t) := (S_t)_{t \geq 0}$  be a family of linear operators in  $\mathcal{L}(E)$  with index  $t \in \mathbb{R}_+$ .  $(S_t)$  is called an **algebraic operator semigroup (AO semigroup)** if  $S_0$  is the identity and  $(S_t)$  has the semigroup property

$$S_{s+t} = S_t \circ S_s \quad \text{for all } s, t \in \mathbb{R}_+.$$

If, in addition,  $t \mapsto S_t u$  is continuous for all  $u \in E$ , then  $(S_t)$  is called a  **$C_0$ -semigroup**. Given  $u \in E$ , the function  $t \mapsto S_t u$  is called the **orbit** or **trajectory** of  $u$  under  $(S_t)$ . The point  $u$  is called the **initial condition**.

**3.2. Continuity Results.** Let  $K$  be a compact subset of  $\mathbb{R}$  and let  $\{S_t\}_{t \in K}$  be a subset of  $\mathcal{L}(E)$ . The following result is from Engel and Nagel (2006).

**Lemma 3.1.** *The following statements are equivalent:*

- (i) *The map  $t \mapsto S_t u$  is continuous on  $K$  for all  $u \in E$ .*
- (ii)  *$\|S_t\|$  is bounded over  $t \in K$  and there exists a dense subset  $D$  of  $E$  such that  $t \mapsto S_t u$  is continuous on  $K$  for all  $u \in D$ .*
- (iii) *For any compact  $C \subset \mathbb{R}$ , the map  $(t, u) \mapsto S_t u$  is uniformly continuous on  $K \times C$ .*

*Proof.* ((i)  $\implies$  (ii)) By (i), for any  $u \in E$ , the map  $t \mapsto S_t u$  is continuous on a compact set and, therefore, its image is bounded in  $E$ . Hence, by the uniform boundedness

principle,  $\|S_t\|$  is bounded over  $t \in K$ . The statement in (ii) regarding continuity is obvious.

((ii)  $\implies$  (iii)). Fix compact  $C \subset E$  and  $\varepsilon > 0$ . We metrize  $K \times C$  by setting  $d((s, u), (t, v)) = \|u - v\| \vee |s - t|$ . Choose  $M \in \mathbb{N}$  such that  $\|S_t\| \leq M$  for all  $t \in K$ . Let  $D$  be the dense set in (ii) and observe that the set of open balls  $B(u, \varepsilon/M)$  over  $u \in D$  provides an open cover of  $C$ . As such, we can choose a finite set  $D_F \subset D$  such that  $C$  is contained in  $\cup_{u \in D_F} B(u, \varepsilon/M)$ . Since, for each  $u \in D_F$ , the map  $t \mapsto S_t u$  is continuous on a compact set, it is also uniformly continuous. As a result, given  $u \in D_F$ , we can select a  $\delta_u > 0$  such that

$$|s - t| < \delta_u \implies \|S_s u - S_t u\| < \varepsilon.$$

Let  $\delta$  be the minimum of  $\{\delta_u\}_{u \in D_F}$  and  $\varepsilon/M$ . If we take  $u, v \in C$  and  $s, t \in K$  with  $d((s, u), (t, v)) < \delta$ , then, choosing  $w \in D_F$  with  $\|u - w\| < \varepsilon/M$ , we have

$$\begin{aligned} \|S_s u - S_t v\| &\leq \|S_s u - S_s w\| + \|S_s w - S_t w\| + \|S_t w - S_t v\| \\ &= \|S_s(u - w)\| + \|S_s w - S_t w\| + \|S_t(w - v)\| \\ &< M(\varepsilon/M) + \varepsilon + M(2\varepsilon/M) = 4\varepsilon. \end{aligned}$$

Hence  $(t, u) \mapsto S_t u$  is uniformly continuous on  $K \times C$ , as claimed.

((iii)  $\implies$  (i)) This implication is trivial (take  $C$  to be a singleton).  $\square$

**Lemma 3.2.** *If  $(S_t)_{t \geq 0}$  is a  $C_0$ -semigroup on  $E$ , then*

- (i)  $\sup_{t \leq \delta} \|S_t\| < \infty$  for all  $\delta > 0$ .
- (ii) there exists an  $a \in \mathbb{R}$  and  $M \geq 1$  such that  $\|S_t\| \leq M \exp(at)$  for all  $t \geq 0$ .

*Proof.* Regarding (i), we first claim there exists an  $\varepsilon > 0$  such that  $\sup_{t \leq \varepsilon} \|S_t\| < \infty$ . Indeed, if no such  $\varepsilon$  exists, then there exists a sequence  $t_n \rightarrow 0$  such that  $\|S_{t_n}\|$  is unbounded. But then, by the principle of uniform boundedness, there exists a  $u \in E$  such that  $\|S_{t_n} u\|$  is unbounded. This contradicts the continuity property of  $C_0$ -semigroups.

Now let  $\varepsilon$  be as above and choose  $M \in \mathbb{N}$  with  $\|S_t\| \leq M$  whenever  $t \leq \varepsilon$ . Fix  $k \in \mathbb{N}$  and  $t \leq k\varepsilon$ . Since  $S_t$  is  $k$  compositions of  $S_{t/k}$ , and since  $t/k \leq \varepsilon$ , the semigroup property yields  $\|S_t\| \leq kM$ . Hence  $t \mapsto S_t$  is bounded on  $[0, k\varepsilon]$ . Since  $k$  was an arbitrary element of  $\mathbb{N}$ , this proves the claim in Lemma 3.2.

The proof of (ii) follows from (i). For details see, for example, Theorem 1.2.8 of [Applebaum \(2019\)](#).  $\square$

**Lemma 3.3.** *If  $(S_t)$  is an AO semigroup on  $E$ , then the following statements are equivalent:*

- (i)  $(S_t)$  is a  $C_0$ -semigroup on  $E$ .
- (ii)  $\lim_{t \downarrow 0} S_t u = u$  for all  $u \in E$ .

*Proof.* That (i) implies (ii) is obvious. For the reverse implication, fix  $u \in E$  and  $t > 0$ . We need to show that  $\|S_{t+h}u - S_t u\| \rightarrow 0$  as  $h \rightarrow 0$ . Suppose first that  $h \downarrow 0$ . Then

$$\|S_{t+h}u - S_t u\| = \|S_t S_h u - S_t u\| \leq \|S_t\| \|S_h u - u\| \rightarrow 0.$$

If, on the other hand  $h \uparrow 0$ , then

$$\|S_{t+h}u - S_t u\| = \|S_{t+h}u - S_{t+h}S_{-h}u\| \leq \|S_{t+h}\| \|u - S_{-h}u\| \rightarrow 0.$$

In the last step we used the fact that  $\|S_{t+h}\|$  is bounded over  $h$  by Lemma 3.2.  $\square$

**Lemma 3.4.** *Let  $(S_t)_{t \geq 0}$  be an AO semigroup on  $E$ . If there exists a dense subset  $D$  of  $E$  such that  $\lim_{t \downarrow 0} S_t u = u$  for all  $u \in D$  and, in addition,  $\sup_{t \leq \delta} \|S_t\| < \infty$  for some  $\delta > 0$ , then  $(S_t)_{t \geq 0}$  is a  $C_0$ -semigroup.*

*Proof.* Fix  $u \in E$ . By Lemma 3.3 it suffices to show that, for a given sequence  $t_n \downarrow 0$ , we have  $S_{t_n}u \rightarrow u$  as  $n \rightarrow \infty$ . To see that this holds, fix  $t_n \downarrow 0$  and choose a compact subset  $K$  of  $\mathbb{R}_+$  such that  $\{t_n\} \subset K$ . Since  $K$  is compact,  $K \ni t \mapsto S_t w$  is continuous when  $w \in D$ , and  $\|S_t\|$  is bounded over  $t \in K$ , Lemma 3.1 implies that  $K \ni t \mapsto S_t u$  is continuous. In particular,  $S_{t_n}u \rightarrow u$  as  $n \rightarrow \infty$ .  $\square$

**3.3. Examples.** We list some examples of semigroups acting on Banach spaces.

**3.3.1. Left-Shift Semigroups.** Let  $C_V(\mathbb{R}_+, \mathbb{R})$  be the set of all continuous real-valued functions  $f$  on  $\mathbb{R}_+$  with  $f(x) \rightarrow 0$  as  $x \rightarrow \infty$ . The set  $C_V(\mathbb{R}_+, \mathbb{R})$  is paired with the supremum norm. Consider the [left translation semigroup](#) given by  $(S_t f)(x) = f(x+t)$ .

**Lemma 3.5.**  $(S_t)$  is a  $C_0$ -semigroup on  $C_V(\mathbb{R}_+, \mathbb{R})$ .

*Proof.* It is clear that  $S_0$  is the identity. The semigroup property holds because, for  $s, t \geq 0$ , we have

$$S_{s+t}f = f(\cdot + s + t) = S_t S_s f$$

Regarding continuity, fix  $f \in C_V(\mathbb{R}_+, \mathbb{R})$  and let  $(t_n)$  be a real sequence with  $t_n \downarrow 0$ . Fix  $\varepsilon > 0$ . Since  $f$  is uniformly continuous, we can select a  $\delta > 0$  such that  $|f(x) - f(y)| < \varepsilon$

whenever  $|x - y| < \delta$ . Let  $N \in \mathbb{N}$  be such that  $t_n < \delta$  when  $n \geq N$ . Then, for  $n \geq N$ ,

$$\|S_{t_n}f - f\| = \sup_x |f(x + t_n) - f(x)| < \varepsilon.$$

Hence  $S_t f \downarrow f$  and  $(S_t)$  is a  $C_0$ -semigroup.  $\square$

Let  $C_0^1(\mathbb{R}_+, \mathbb{R})$  be the set of all continuously differentiable  $f \in C_V(\mathbb{R}_+, \mathbb{R})$  with  $f' \in C_V(\mathbb{R}_+, \mathbb{R})$ . The set  $C_0^1(\mathbb{R}_+, \mathbb{R})$  is paired with the norm  $\|f\| = \sup_x |f(x)| + \sup_x |f'(x)|$ .

**Lemma 3.6.**  $(S_t)$  is a  $C_0$ -semigroup on  $C_0^1(\mathbb{R}_+, \mathbb{R})$ .

*Proof.* In view of Lemma 3.5, we only need to check continuity. Fixing  $f \in C_0^1(\mathbb{R}_+, \mathbb{R})$ , we have

$$\|S_t f - f\| = \sup_x |f(x + t) - f(x)| + \sup_x |f'(x + t) - f'(x)|$$

Since  $f$  and  $f'$  are both in  $C_V(\mathbb{R}_+, \mathbb{R})$ , the proof of Lemma 3.5 implies that both terms on the right hand side converge to zero as  $t \downarrow 0$ . Hence continuity holds.  $\square$

**3.3.2. Right-Shift Semigroups.** Here we discuss right-shift semigroups. We will embed them in a space of integrable functions. Below  $\lambda$  denotes Lebesgue measure.

Let  $C_K(\mathbb{R})$  be the set of all continuous real-valued functions  $f$  on  $\mathbb{R}$  that vanish off a compact set. Let  $L_1(\mathbb{R})$  be the set of Borel measurable real-valued functions on  $\mathbb{R}$  with  $\|f\| := \int |f| d\lambda < \infty$ . Let  $S_t$  be the linear operator on  $L_1(\mathbb{R})$  defined by

$$(S_t f)(x) = f(x - t) \quad (x \in \mathbb{R}, t \geq 0).$$

**Lemma 3.7.**  $(S_t)$  is a  $C_0$ -semigroup on  $L_1(\mathbb{R})$ .

*Proof.* It is simple to confirm that  $(S_t)$  is an AO semigroup on  $L_1(\mathbb{R})$ . Regarding continuity, note that  $\|S_t f\| = \int |f(x - t)| dx = \|f\|$ , so  $\|S_t\|$  is bounded in  $t$ . Since  $C_K(\mathbb{R})$  is dense in  $L_1(\mathbb{R})$  under this norm, Lemma 3.4 implies that, to show  $(S_t)$  is a  $C_0$ -semigroup on  $L_1(\mathbb{R})$ , it suffices to show that  $\|S_t f - f\| \rightarrow 0$  for any  $f \in C_K(\mathbb{R})$ .

To this end, fix  $f \in C_K(\mathbb{R})$  and let  $K$  be a compact set such that  $f$  vanishes off  $K$ . Fix  $\varepsilon > 0$ . By uniform continuity, we can take a  $\delta > 0$  such that  $|x - y| < \delta$  implies  $|f(x) - f(y)| < \varepsilon/\lambda(K)$ . If  $t < \delta$ , then

$$\|S_t f - f\| = \int |f(x - t) - f(x)| dx \leq \lambda(K) \frac{\varepsilon}{\lambda(K)} = \varepsilon.$$

This completes the proof of  $C_0$ -continuity of  $(S_t)$  on  $L_1(\mathbb{R})$ .  $\square$

3.3.3. *Multiplication Semigroups.* Let  $(X, \mathcal{B}, \mu)$  be a  $\sigma$ -finite measure space and let  $\varphi$  be a measurable map from  $X$  to  $\mathbb{R}_+$ . Define

$$(S_t f)(x) = \exp(-t\varphi(x))f(x) \quad (x \in X, t \geq 0).$$

The family  $(S_t)$  is called a [multiplication semigroup](#).

**Lemma 3.8.**  $(S_t)$  is a  $C_0$ -semigroup on  $L_1(X, \mathcal{B}, \mu)$ .

*Proof.* It is simple to confirm that  $(S_t)$  is an AO semigroup on  $L_1(\mathbb{R})$ . Regarding continuity, fix  $f \in L_1(X, \mathcal{B}, \mu)$  and observe that

$$\|S_t f - f\| = \int |f(x)| |\exp(-t\varphi(x)) - 1| \mu(dx).$$

It follows from the dominated convergence theorem that this integral converges to zero as  $t \downarrow 0$ . Hence  $(S_t)$  is a  $C_0$ -semigroup on  $L_1(X, \mathcal{B}, \mu)$ .  $\square$

3.3.4. *Uniformly Continuous Semigroups.* Fixing  $A \in \mathcal{L}(E)$ , consider the family of linear operators on  $E$  given by

$$S_t u = \exp(tA)u \quad (u \in E, t \geq 0)$$

We recall that the exponential function  $\varphi(t) := \exp(tA)$

- (i) obeys  $\varphi(0) = I$  and  $\varphi(s+t) = \varphi(t)\varphi(s)$  for all  $s, t \in \mathbb{R}$ ; and
- (ii) is continuous as a map from  $\mathbb{R}$  to  $\mathcal{L}(E)$ .

From (i) we can easily confirm that  $(S_t)$  is an algebraic operator semigroup on  $E$ . Regarding continuity, (i) and (ii) imply that

$$\lim_{t \downarrow 0} \|S_t - I\| = 0. \tag{7}$$

It follows from (7) that  $(S_t)$  is a  $C_0$ -semigroup on  $E$ .

Any operator semigroup  $(S_t)$  on  $E$  obeying (7) is called a [uniformly continuous semigroup](#). In fact no other examples exist:

**Theorem 3.9.** *If  $(S_t)$  is a uniformly continuous semigroup on  $E$ , then there exists an  $A \in \mathcal{L}(E)$  such that  $S_t u = \exp(tA)u$  for all  $u \in E$  and  $t \geq 0$ .*

The proof of Theorem 3.9 can be found 2(b) of [Engel and Nagel \(2006\)](#).

## 4. GENERATORS AND RESOLVENTS

Add roadmap.

**4.1. Infinitesimal Generators.** Let  $E$  be a Banach space and let  $(S_t)$  be an AO semigroup on  $E$ . Set

$$\mathcal{D}(A) = \left\{ u \in E \text{ s.t. } \lim_{t \downarrow 0} \frac{S_t u - u}{t} \text{ exists} \right\} \quad \text{and} \quad Au = \lim_{t \downarrow 0} \frac{S_t u - u}{t} \text{ on } \mathcal{D}(A).$$

The map  $A$  is called the **infinitesimal generator** (or just **generator**) of  $(S_t)$  and  $\mathcal{D}(A)$  is its domain. We sometimes also refer to the pair  $(A, \mathcal{D}(A))$  as the generator.

**Lemma 4.1.** *The set  $\mathcal{D}(A)$  is a linear subspace of  $E$  and  $A$  is a closed linear operator from  $\mathcal{D}(A)$  to  $E$ .*

*Proof.* Fix  $\alpha, \beta \in \mathbb{R}$  and  $u, v \in \mathcal{D}(A)$ . Let  $w = \alpha u + \beta v$ . Since

$$\frac{S_t w - w}{t} = \alpha \frac{S_t u - u}{t} + \beta \frac{S_t v - v}{t},$$

we see that  $w \in \mathcal{D}(A)$  and, moreover,  $Aw = \alpha Au + \beta Av$ . This proves that  $A$  is a linear operator from linear subspace  $\mathcal{D}(A)$  to  $E$ . The proof that  $A$  is closed can be found in Theorem 1.3.16 of [Applebaum \(2019\)](#).  $\square$

The next result shows that any trajectory of  $(S_t)$  starting from a point  $\bar{u}$  in  $\mathcal{D}(A)$  obeys a linear differential equation evolving in  $\mathcal{D}(A)$ , with  $A$  as the “vector field.”

**Lemma 4.2.** *If  $(S_t)$  is a  $C_0$ -semigroup and  $u \in E$ , then, for all  $t \geq 0$ ,*

$$\int_0^t S_s u \, ds \in \mathcal{D}(A) \quad \text{and} \quad S_t u = u + A \int_0^t S_s u \, ds. \quad (8)$$

*Proof.* Fix  $u \in E$  and  $t \geq 0$ . For  $h > 0$ , the properties of the integral described in Lemma 2.1 imply that

$$\begin{aligned} S_h \int_0^t S_s u \, ds - \int_0^t S_s u \, ds &= \int_0^t S_{h+s} u \, ds - \int_0^t S_s u \, ds \\ &= \int_h^{t+h} S_s u \, ds - \int_0^t S_s u \, ds = \int_t^{t+h} S_s u \, ds - \int_0^h S_s u \, ds \end{aligned}$$

Multiplying by  $(1/h)$  and taking the limit as  $h \downarrow 0$  gives  $A \int_0^t S_s u \, ds = S_t u - u$ . Hence both claims in (8) hold.  $\square$

**Proposition 4.3.** *If  $(S_t)$  is a  $C_0$ -semigroup, then  $\mathcal{D}(A)$  is dense in  $E$ . Moreover, for any  $u \in \mathcal{D}(A)$  and  $t \geq 0$ ,*

$$(i) \quad S_t u \in \mathcal{D}(A),$$

(ii) the map  $s \mapsto S_s u$  is differentiable on  $\mathbb{R}_+$  and

$$\frac{d}{dt} S_t u = A S_t u = S_t A u, \quad \text{and} \quad (9)$$

(iii)  $S_t u = u + \int_0^t S_s A u \, ds$  for all  $u \in \mathcal{D}(A)$  and  $t \geq 0$ .

*Proof.* To see that  $\mathcal{D}(A)$  is dense, consider  $v_h := (1/h) \int_0^h S_s u \, ds$  with  $h \geq 0$ . By Lemma 2.1,  $v_h \rightarrow u$  as  $h \downarrow 0$ . Moreover, (8) implies that  $v_h$  is in  $\mathcal{D}(A)$  for all  $h > 0$ . Hence  $\mathcal{D}(A)$  is dense in  $E$ .

Regarding (i), fix  $t \geq 0$  and  $u \in \mathcal{D}(A)$ . We have

$$A S_t u = \lim_{h \downarrow 0} \frac{S_h S_t u - S_t u}{h} = \lim_{h \downarrow 0} \frac{S_t S_h u - S_t u}{h} = \lim_{h \downarrow 0} S_t \frac{S_h u - u}{h} = S_t A u,$$

where the last equality is by continuity of  $S_t$  (i.e.,  $S_t \in \mathcal{L}(E)$ ). This proves both (i) and the second equality in (9).

We now show that  $s \mapsto S_s u$  is differentiable at  $t \in \mathbb{R}_+$  with derivative  $S_t A u$ , which will complete the proof of (ii). We choose  $t > 0$  because the case  $t = 0$  is trivial. The right derivative of  $S_t u$  is

$$\lim_{h \downarrow 0} \frac{S_{t+h} u - S_t u}{h} = \lim_{h \downarrow 0} \frac{S_t S_h u - S_t u}{h}$$

which has already been shown to equal  $S_t A u$ .

Regarding the left-hand derivative, we take  $0 < h < t$  and use

$$\frac{S_{t-h} u - S_t u}{-h} - S_t A u = S_{t-h} \left( \frac{u - S_h u}{-h} - S_h A u \right)$$

and the boundedness of  $\|S_t\|$  over bounded sets (Lemma 3.2) to obtain a finite  $M$  such that

$$\left\| \frac{S_{t-h} u - S_t u}{-h} - S_t A u \right\| \leq M \left\| \frac{u - S_h u}{-h} - A u \right\| + M \|A u - S_h A u\|.$$

Both terms on the right converge to zero in  $h$ , so we have proved that the left derivative is also to  $S_t A u$ . This completes the proof of (ii).

Regarding (iii), let  $\varphi(s) = S_s u$  for all  $s \geq 0$ . We have shown in (ii) that  $\varphi$  is differentiable on  $\mathbb{R}_+$  with  $\dot{\varphi}(s) = S_s A u$ . With this notation, the claim in (iii) can be expressed as  $\varphi(t) = \varphi(0) + \int \dot{\varphi}(s) \, ds$ . Since  $s \mapsto S_s A u$  is continuous in  $s$ ,  $\varphi$  is also continuously differentiable. The claim in (iii) now follows from (2).  $\square$

The next result shows that semigroups are uniquely identified by their generators.

**Proposition 4.4.** *Let  $(S_t)$  and  $(T_t)$  be two  $C_0$ -semigroups on  $E$  with generators  $(A, \mathcal{D}(A))$  and  $(B, \mathcal{D}(B))$ . If  $(A, \mathcal{D}(A)) = (B, \mathcal{D}(B))$ , then  $(S_t) = (T_t)$ .*

*Proof.* Let the statement hold, so that  $(S_t)$  and  $(T_t)$  are two  $C_0$ -semigroups with generator  $A$  on domain  $\mathcal{D}(A)$ . Fix  $t > 0$ . We claim that  $S_t = T_t$  on  $E$ . Since  $\mathcal{D}(A)$  is dense in  $E$ , it suffices to show that  $S_t$  and  $T_t$  agree on  $\mathcal{D}(A)$ .

To this end, fix  $u \in \mathcal{D}(A)$  and let  $s \mapsto v(s)$  be a differentiable function on  $\mathbb{R}_+$  with  $\dot{v} = Av$  and  $v(0) = u$ . Define  $w(s) := S_{t-s}v(s)$  for all  $s \in (0, t)$ . For  $h$  close to zero we have

$$\begin{aligned} w(s+h) - w(s) &= S_{t-s-h}v(s+h) - S_{t-s}v(s) \\ &= S_{t-s-h}(v(s+h) - v(s)) + S_{t-s-h}v(s) - S_{t-s}v(s) \end{aligned}$$

Dividing by  $h$  gives

$$\frac{w(s+h) - w(s)}{h} = S_{t-s-h} \frac{v(s+h) - v(s)}{h} - S_{t-s} \frac{S_{-h}v(s) - v(s)}{-h} \quad (10)$$

Consider the first term in (10). Letting

$$d(h) := \left\| S_{t-s-h} \frac{v(s+h) - v(s)}{h} - S_{t-s}Av(s) \right\|$$

and, choosing  $M$  suitably large, we have

$$\begin{aligned} d(h) &\leq \|S_{t-s-h}\| \left\| \frac{v(s+h) - v(s)}{h} - S_hAv(s) \right\| \\ &\leq M \left\| \frac{v(s+h) - v(s)}{h} - Av(s) \right\| + M \|Av(s) - S_hAv(s)\| \end{aligned}$$

This confirms that  $d(h) \rightarrow 0$  as  $h \rightarrow 0$ , which means that the first term in (10) converges to  $S_{t-s}Av(s)$ .

The second term also converges to  $S_{t-s}Av(s)$ . Hence  $w$  is differentiable on  $(0, t)$  with  $\dot{w} = 0$ . It follows from (2) that  $w(s) = w(0) = S_t u$  for all  $s < t$ . Hence  $S_{t-s}v(s) = S_t u$  for all  $s < t$ . Taking  $s \rightarrow t$  and applying continuity gives  $v(t) = S_t u$ .

To complete the proof we consider the case  $v(s) = T_s u$ . According to Proposition 4.3, the function  $v$  is differentiable on  $\mathbb{R}_+$  with  $\dot{v} = Av$  and  $v(0) = v$ . Hence  $T_t u = S_t u$ . This confirms that  $S_t$  and  $T_t$  agree on  $\mathcal{D}(A)$ .  $\square$

Let's now translate the results above in to findings for IVPs with potentially unbounded generators. We confirm that many of the results from the bounded case (see §2.3) either carry over or have direct analogs with the general case.

In the next theorem,  $C_1(\mathbb{R}_+, E)$  is the set of continuously differentiable functions from  $\mathbb{R}_+$  to  $E$ .

**Theorem 4.5.** *Let  $(S_t)$  be a  $C_0$ -semigroup with generator  $(A, \mathcal{D}(A))$  and let  $u_0$  be a point in  $\mathcal{D}(A)$ . The unique solution in  $C(\mathbb{R}_+, E)$  of the initial value problem  $\dot{u} = Au$  with  $u(0) = u_0$  is the function defined by  $u(t) := S_t u_0$ . Moreover,  $u(t)$  is continuously differentiable on  $\mathbb{R}_+$  and*

$$\dot{u}(t) = AS_t u = S_t A u_0.$$

*Proof.* Most of the claims in Theorem 4.5 follow directly from Proposition 4.3. The only missing components is uniqueness. To prove it, fix  $u_0 \in \mathcal{D}(A)$  and let  $v$  be a differentiable function from  $\mathbb{R}_+$  to  $E$  with  $\dot{v}(t) = Av(t)$  for all  $t$  and  $v(0) = u_0$ . We already showed in the proof of Proposition 4.4 that for any such function we have  $v(t) = S_t u_0 = u(t)$ . Hence uniqueness also holds.  $\square$

**4.2. Resolvents.** Let  $(A, \mathcal{D}(A))$  be the generator of a  $C_0$ -semigroup  $(S_t)$  on  $E$ . Recalling that  $A$  is a closed linear operator (Lemma 4.1), the **resolvent** of  $A$  is, for each  $\lambda \in \rho(A)$ , the bounded and bijective linear operator

$$R_\lambda(A) = (\lambda - A)^{-1}$$

from  $E$  to  $\mathcal{D}(A)$ . We also refer to  $R_\lambda(A)$  as the resolvent of the semigroup  $(S_t)$ . We write  $R_\lambda$  instead of  $R_\lambda(A)$  when  $A$  is understood.

**Proposition 4.6.** *Let  $A$  be the generator of a  $C_0$ -semigroup  $(S_t)$  satisfying  $\|S_t\| \leq Me^{at}$  for all  $t \geq 0$ . If  $\lambda \in \mathbb{C}$  with  $\Re(\lambda) > a$ , then  $\lambda$  is in the resolvent set  $\rho(A)$ . Moreover,*

(i) *the operator norm of the resolvent obeys*

$$\|R_\lambda\| \leq \frac{M}{\Re(\lambda) - a} \quad \text{and}$$

(ii) *the resolvent has the integral representation*

$$R_\lambda = \int_0^\infty e^{-\lambda t} S_t \, dt.$$

*Proof.* Let  $(S_t)$  be as stated, with  $\|S_t\| \leq M \exp(at)$  for all  $t \geq 0$ . Fix  $\lambda \in \mathbb{C}$  with  $\Re(\lambda) > a$ . Let  $T_\lambda = \int_0^\infty e^{-\lambda t} S_t \, dt$ . Fixing  $u \in E$ , we have

$$\|T_\lambda u\| = \left\| \int_0^\infty e^{-\lambda t} S_t u \, dt \right\| \leq \int_0^\infty |e^{-\lambda t}| \cdot \|S_t u\| \, dt$$

Since  $|e^{-\lambda t}| = |e^{-\Re(\lambda)t}|$ , we obtain

$$\|T_\lambda u\| \leq \|u\| M \int_0^\infty e^{(a-\Re(\lambda))t} \, dt = \frac{M}{\Re(\lambda) - a} \|u\|.$$

This proves that  $T_\lambda$  is well-defined and bounded on  $E$ , and that (i) holds. Next we claim that  $T_\lambda(\lambda - A) = (\lambda - A)T_\lambda = I$ , which is equivalent to

$$T_\lambda Au = AT_\lambda u = \lambda T_\lambda u - u \quad \text{for all } u \in E. \quad (11)$$

Since  $T_\lambda$  and  $(S_t)$  are bounded,

$$\begin{aligned} T_\lambda Au &= \int_0^\infty e^{-\lambda t} S_t \lim_{h \downarrow 0} \frac{S_h - I}{h} u \, dt \\ &= \lim_{h \downarrow 0} \int_0^\infty e^{-\lambda t} S_t \frac{S_h - I}{h} u \, dt = \lim_{h \downarrow 0} \frac{S_h - I}{h} \int_0^\infty e^{-\lambda t} S_t u \, dt = AT_\lambda u \end{aligned}$$

This establishes the first equality in (11). Regarding the second, observe that

$$\begin{aligned} AT_\lambda u &= \lim_{h \downarrow 0} \frac{S_h - I}{h} \int_0^\infty e^{-\lambda t} S_t u \, dt = \lim_{h \downarrow 0} \left( \frac{1}{h} \int_0^\infty e^{-\lambda t} S_{t+h} u \, dt \right) - \frac{1}{h} \int_0^\infty e^{-\lambda t} S_t u \, dt \\ &= \lim_{h \downarrow 0} \left( \frac{1}{h} \int_h^\infty e^{-\lambda(t-h)} S_t u \, dt \right) - \frac{1}{h} \int_0^\infty e^{-\lambda t} S_t u \, dt \\ &= \lim_{h \downarrow 0} -e^{\lambda h} \frac{1}{h} \int_0^h e^{-\lambda t} S_t u \, dt + \lim_{h \downarrow 0} \frac{1}{h} (e^{\lambda h} - 1) \int_0^\infty e^{-\lambda t} S_t u \, dt = -u + \lambda T_\lambda u \end{aligned}$$

It follows that  $\lambda$  is in the resolvent set  $\rho(A)$  and that the integral representation in (ii) holds.  $\square$

**4.3. Example: Diffusion Semigroups.** Consider the Banach Space  $E = C[0, 1]$  and the differential operator

$$Af := f''$$

with the domain  $D(A) := \{f \in C^2[0, 1] : f'(0) = f'(1) = 0\}$ . Now we define  $(e_n)$  as

$$s \mapsto e_n(s) := \begin{cases} 1 & \text{if } n = 0 \\ \sqrt{2} \cos(\pi ns) & \text{if } n \geq 1 \end{cases}$$

It is obvious that  $(e_n) \subset D(A)$ .

**Lemma 4.7.**  $Y := \text{lin}\{e_n : n \geq 0\}$  is a dense algebra of  $E$ .

*Proof.* By Stone–Weierstrass theorem, The set of all polynomial functions forms a dense subalgebra of  $E$ . That is,

$$\text{lin}\{f_n : f_n(s) = s^n, s \in [0, 1], n \geq 0\} \quad (12)$$

is a dense subalgebra of  $E$ . We now let  $s = \cos(\pi x)$  with  $x \in [0, 1]$  and so  $s \in [0, 1]$ . In this way, we replace  $s$  in (12) by  $\cos(\pi x)$  and then every  $f \in E$ ,  $f(s) = f(\cos(x))$  for all  $s \in [0, 1]$  with  $s = \cos(x)$ . Then we have

$$Y' := \text{lin}\{f_n : f_n(s) = \cos(\pi s)^n, s \in [0, 1], n \geq 0\}$$

is a dense subalgebra of  $E$ .

By trigonometric identities, we have

$$\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos(\alpha) \cos(\beta) \quad (13)$$

Thus, by induction,  $\cos(\pi s)^n \in Y$  for all  $n$  and so  $Y' \subset Y$  which finishes the proof.  $\square$

As we know that  $D(A)$  is a linear space containing  $(e_n)$ ,  $Y \subset D(A)$  and so  $D(A)$  is a dense subspace of  $E$ . Define the [rank-one operators](#) as

$$e_n \otimes e_n : f \mapsto \langle f, e_n \rangle e_n := \left( \int_0^1 f(s) e_n(s) ds \right) e_n$$

where  $\|e_n \otimes e_n\| \leq 2$  and  $(e_n \otimes e_n) e_m = \mathbf{1}_{\{m=n\}} e_n$  for all  $m, n$ . The operators  $(S_t)$  is defined as

$$S_t = \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot e_n \otimes e_n \quad (14)$$

Then we have for  $f \in E$ ,

$$\begin{aligned} S_t f &= \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot \langle f, e_n \rangle e_n \\ &= \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot \left( \int_0^1 f(s) e_n(s) ds \right) e_n \end{aligned}$$

For  $r \in [0, 1]$ ,

$$\begin{aligned} (S_t f)(r) &= \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot \left( \int_0^1 f(s) e_n(s) ds \right) e_n(r) \\ &= \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot \left( \int_0^1 f(s) e_n(s) e_n(r) ds \right) \\ &= \int_0^1 f(s) \sum_{n=0}^{\infty} e^{-\pi^2 n^2 t} \cdot e_n(s) e_n(r) ds \\ &= \int_0^1 f(s) \left( 2 \sum_{n=1}^{\infty} e^{-\pi^2 n^2 t} \cdot \cos(n\pi s) \cos(n\pi r) + 1 \right) ds \end{aligned}$$

We denote  $k_t(s, r) = 2 \sum_{n=1}^{\infty} e^{-\pi^2 n^2 t} \cdot \cos(n\pi s) \cos(n\pi r) + 1$ , and  $(S_t f)(r) = \int_0^1 f(s) k_t(s, r) ds$ . For all  $S_t$ ,

$$\|S_t\| = \sup_{f \in E} \frac{\|S_t f\|}{\|f\|} = \sup_{f \in E} \frac{\sup_{r \in [0,1]} \left\| \int_0^1 f(s) k_t(s, r) ds \right\|}{\|f\|} \leq \sup_{r \in [0,1]} \left\| \int_0^1 k_t(s, r) ds \right\| = 1$$

and since  $\|S_t \mathbb{1}\| = 1$ ,  $\|S_t\| = 1$ .

The Jacobi identity is

$$w_t(s) = \frac{1}{\sqrt{4\pi t}} \sum_{\mathbb{Z}} e^{-(s+2n)^2/4t} = \frac{1}{2} + \sum_{n=1}^{\infty} e^{-\pi^2 n^2 t} \cdot \cos(n\pi s)$$

Applying (13),  $k_t(s, r) = w_t(s+r) + w_t(s-r)$ .

**Proposition 4.8.**  $(S_t)$  with  $S_0 = I$  is a  $C_0$  semigroup on  $E$  and its generator is given by

$$Af = f''$$

with the domain  $D(A) := \{f \in C^2[0, 1] : f'(0) = f'(1) = 0\}$ .

*Proof.* Note that the property of rank-one operators  $(e_n \otimes e_n) e_m = \mathbf{1}_{\{m=n\}} e_n$  for all  $m, n$ . It is obvious that  $S_{t+s} = S_t S_s$  holds on  $Y$  and so on  $E$  by continuity. Similarly,  $t \mapsto S_t u$  is continuous for all  $u \in E$ .

We next show the generator of  $B$  of  $(S_t)$  coincides with  $A$  defined above. By simple calculations,  $Bf = Af$  for all  $f \in Y$ . By definition of  $A$  and Wiki<sup>1</sup>,  $1 \in \rho(A)$ . Theorem 1.10(ii) of Engel and Nagel (2006) implies  $1 \in \rho(B)$  as well. Thus,  $(A - I)^{-1}$  coincides with  $(B - I)^{-1}$  on  $B(Y)$  which is dense in  $E$ . Boundedness of  $(A - I)^{-1}$  and  $(B - I)^{-1}$  implies  $(A - I)^{-1}$  and  $(B - I)^{-1}$  agree on  $E$  and so  $A = B$ .  $\square$

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<sup>1</sup>[https://en.wikipedia.org/wiki/Spectral\\_theory\\_of\\_ordinary\\_differential\\_equations](https://en.wikipedia.org/wiki/Spectral_theory_of_ordinary_differential_equations)

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